Staff answers and design checks

Design and Technology LAUNCH Spring 2 Week 1 Thursday 25 February 2027 P1 Year A

The problem and specification

Sam is invented and needs one 80 mm wide by 60 mm high paper card upright and readable. A useful specification names card support and message view, stability on a level table for 30 seconds with hands away, safe handling and removal by hand. The overall table area is 160 mm by 120 mm including the projecting channel. The triangular body base alone is 120 mm by 80 mm. These are invented targets, not standards.

Three ideas

Accept three distinct pupil-generated or pupil-directed ideas. The prompt models are folded triangle with a U channel, slot base with upright backplate, and upright back with triangular braces. Pupils may change them or invent another feasible idea. Three ticked model names alone are not three annotated pupil ideas.

Useful annotations

Each idea has at least two labels naming a part and its useful job. Adult example: broad base may help stability; a U channel may catch the card's lower edge. Colour can be a chosen feature but cannot replace a structural reason. A pupil may explain labels orally with confirmed scribing.

Own choice

Accept a justified choice linked to the same specification. One design is not automatically correct. The adult checks practical feasibility later. Record two reasons and one question or actual change; do not manufacture an improvement or test result.

Default model details

If asked: the planned main blank is 120 mm by 260 mm, folded at 80, 160 and 240 mm from one end; the 20 mm tab closes the triangle with cold paper tape. The retaining strip is 100 mm by 30 mm with long folds at 10 and 20 mm, making a U channel. Those are Week 2 planning details, not cutting permission today.

Staff misconceptions and questions

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Hinge and response

Hinge: no, blue colour does not prove the stand will hold the card. Re-point to the problem and name a support feature. Parallel: no, a drawn U channel has not passed an actual slip test. Reteach feature, effect, later check; ask the pupil to explain before moving on.

Drawing quality and ownership

Quick sketches may be rough. Assess distinct solutions, useful annotations and reasons. Do not require artistic skill. If an adult draws, the pupil directs and confirms the design; record that help. A copied adult model cannot be claimed as an independently generated design.

Dimensions and future making

Written measurements govern later construction. Illustrations are not sized templates. An alternative concept or size needs an adult-approved revised cutting list and safe method. No hot glue, saws, drills or power tools are introduced as the default route.

Cold call with response choice

Foundation: What is Sam's problem? Expected: the card falls flat and needs support to be readable.

Middle: Why might a wide base help? Expected: it may help stability on the table; test after making.

Higher: Why is a drawn design not a passed test? Expected: it predicts an effect, while the actual card and made stand still need a cleared test.

Evidence gaps

Unfinished ideas remain unfinished. Record a next step before approving a build plan. No actual practical skill is inferred from a drawing, spoken plan or adult model. The unit award is separate from this lesson evidence.